Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/16/2005
PI Name:	Walker, Thad G Ph.D.		
Project Title:	All-optical high density atom sources		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Madison	**State:**	WI
Zip Code:	53706	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		

Our approach to producing high density, high phase space density atoms is to laser cool atoms confined in a holographically produced far-off-resonance lattice. This Holographic Atom Trap (HAT) is produced by imaging a hologram onto a cloud of laser-cooled atoms. The HAT produces strongly confining forces on atoms in a 10-100 micron volume. Once in the HAT, the atoms will be further cooled using lasers to temperatures of a few micro-Kelvin.
The ultimate densities and phase space densities that can be reached in the HAT depend on the collisional properties of the trap and heating due to photon scattering. Both effects are minimized in the HAT by optical pumping into dark states. We will investigate the possibility of direct laser cooling to quantum degeneracy in the HAT.

We seek to use the HAT as a source for studying quantum transport through atom waveguides. Our waveguide will be formed by a laser beam with a hole down the middle. Characteristic steps in the transmission of the guide as a function

Task Description:	of energy are quantized in units of the DeBroglie wavelength. The small energy scales involved mean that gravity is a substantial complication, and later manifestations of the experiment will be substantially improved in a microgravity environment. We seek to develop new high density, high phase space density sources of cold atoms for scientific exploitation of microgravity environments. Such sources potentially have inherent advantages as compared to currently used technology, namely speed, atom efficiency, and power consumption. We propose to develop and study the properties of these sources, and to use them to address a fundamental physics problem ideally suited to microgravity: quantization of atom conductance through an atom waveguide. Development of high-density atom sources by laser cooling will have advantages over current technology in terms of speed, efficiency, and power consumption. These new sources may be of considerable interest to a wide variety of experiments using ultracold atoms on the ground and in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	For the past year we have honed our high density atom trapping techniques using the Holographic Atom Trap we have demonstrated in previous years. We use forced evaporation to increase the phase space density to near unity, then recompress the trap to either optimize density or minimize cloud size. The densities we have attained are in excess of 10^13 atoms/cc, a record for incoherent cold atom clouds and comparable to densities found in BECs. The evaporation is very rapid, about a second, so the clouds are produced very quickly. The high densities are deduced using three methods, involving measuring the trap numbers, spring constants, and cloud sizes, and also by measuring three-body recombination rates and deducing the densities using previously measured three-body rate coefficients. In order to produce very small clouds, (6 microns in the longest dimension), we cool to very low temperatures (few hundreds of nanoKelvin) and then do the recompression. This results in a cloud size reduction of 3 to 4, giving in turn the very small clouds. These small clouds are ideal for single quantum excitations using Rydberg atoms. This work has been accepted for publication as a Rapid Communication in Physical Review A. We are beginning to study the production of Rydberg atoms in cold atom clouds using coherent two-photon excitation. We have built a frequency-doubled diode laser system that produces 50 mW of light at 480 nm. We have excited atoms to principal quantum numbers in the range from 30-70 using this system. We observe the Rydberg atoms in trap-loss, a method that turns out to be very sensitive. We have been able to perform two-photon Rydberg excitation with a bandwidth of about 2.5 MHz, although we do not know what limits this linewidth at the present time. In addition, we have been able to increase the Rabi frequency to several MHz. Thus we are in a regime where coherent Rabi oscillations should be observable.
Bibliography Type:	Description: (Last Updated: 11/10/2009)
Articles in Peer-reviewed Journals	Walker TG, Saffman M. "Zeros of Rydberg–Rydberg Föster interactions." Journal of Physics B: Atomic, Molecular and Optical Physics. 2005 Jan;38(2):S309-19. http://dx.doi.org/10.1088/0953-4075/38/2/022 , Jan-2005
Articles in Peer-reviewed Journals	Sebby-Strabley J, Newell RTR, Day JO, Brekke E, Walker TG. "High-density mesoscopic atom clouds in a holographic atom trap." Physical Review A 2005 Feb;71(2):021401. http://dx.doi.org/10.1103/PhysRevA.71.021401 , Feb-2005
Articles in Peer-reviewed Journals	Saffman M, Walker TG. "Entangling single and N atom qubits for fast quantum state detection and transmission." Physical Review A 2005 Oct 1;72(4):042302. http://dx.doi.org/10.1103/PhysRevA.72.042302 , Oct-2005